MILITARY SPECIFICATION

RECEIVER, PROGRAMMABLE SCANNING COMMUNICATIONS

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the operational characteristics and test and acceptance requirements for a Programmable Scanning Communications Receiver, hereinafter referred to as the equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of Department of Defense Index of Specifications and Standard (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-94	Resistor, Variable, Composition, General Specification For
MIL-S-901	Shock Tests, High-Impact (HI) Shipboard Machinery, Equipment and Systems, Requirements For (Navy)
MIL-C-915	Cable and Cord Electrical, For Shipboard Use, General Specification For
MIL-E-16400	Electronic, Interior Communications and Navigation Equipment, Naval Ship and Shore, General Specification For
MIL-W-16878	Wire, Electrical, Insulated, General Specification For
MIL-E-17555	Electronic and Electrical Equipment, Accessories, And Repair Parts, Packaging and Packing of
MIL-F-19207	Fuseholder, Extractor Post Type, Blown Fuse Indicating And Non-Indicating, General Specification For
MIL-E-21981	Electronics Type Designations, Identification Plates and Markings, Requirements For
MIL-C-24056	Cabinet, Electronic Equipment, CY-4516(A)/s and CY-4516(B)/s (Single Bay) (Naval Shipboard Use)
MIL-P-28809	Printed Wiring Assemblies
MIL-M-38510	Microcircuit, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ELEX 8111, Washington, D.C. 20363 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification For
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables For Inspection by Attributes
MIL-STD-108	Definitions Of and Basic Requirements For Enclosure For Electric and Electronic Equipment
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-167-1 (SHIPS)	Mechanical Vibrations Of Shipboard Equipment (Type I-Environmental And Type II-Internally Excited)
MIL-STD-188	Military Communication Systems Technical Standards
MIL-STD-242	Electronic Equipment Parts Selected Standards
MIL-STD-415	Test Provisions For Electronic Systems and Associated Equipment, Design Criteria For
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements For The Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-883	Test Methods and Procedures For Microelectronics
MIL-STD-965	Parts Control Program
MIL-STD-1399 Section 300	Interface Standard For Shipboard Systems (Electrical Power, Alternating Current (Metric))
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities
MIL-STD-1679(NAVY)	Weapon System Software Development

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2.1.2 Other Government documents. The following Government documents form a part of this specification to the extent specified herein.

TADSTAND A	Standard Definitions for Embedded Computer Resources in Tactical Digital Systems
TADSTAND B	Standard Embedded Computers, Displays, Peripherals, and Input/Output Devices
TADSTAND C	Computer Programming Language Standardization Policy for Tactical Digital Systems
TADSTAND D	Reserve Capacity Requirements for Tactical Digital Systems

(Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other standards. The following document forms a part of this specification to the extent specified herein.

ANSI/EIA AS-310-C-77	Racks, Panels, and Associated Equipment

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The equipment shall be produced in accordance with the provisions of MIL-E-16400, to the extent specified herein. The equipment shall be a digitally tuned super-heterodyne receiver with programmable scanning throughout the frequency range 2.0 to 30.0 MHz. It shall provide reception of single sideband (SSB), upper sideband (USB), lower sideband (LSB), and independent sideband (ISB), frequency modulation (FM), continuous wave (CW), amplitude modulation (AM), and, with an external converter, Frequency-Shift Keyed (FSK) single channel teletypewriter.

3.1.1 First article. Sample units provided for first article inspection shall be suitable for complete evaluation of mechanical and electrical form, design, and performance. They shall be of final mechanical and electrical form, employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of final production equipment. (See 4.3 and 6.2.)

3.1.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with MIL-E-16400, to the extent specified herein. Particular attention shall be given to the provisions of MIL-E-16400 concerning acceptable and unacceptable materials.

3.1.2.1 Parts control. The equipment parts furnished in accordance with MIL-STD-242 shall employ standard parts or approved nonstandard parts as prescribed in MIL-STD-965, Procedure 1.

3.1.2.2 Individual selection. The performance of the equipment shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity.

3.1.2.3 Microelectronic devices. Selection and application of microelectronic devices shall comply with Requirement 64 of MIL-STD-454, requiring MIL-M-38510 Class B devices. When deviations from these requirements are necessitated due to availability or cost considerations, hermetically sealed devices, inspected and tested in accordance with the Non-JAN devices paragraph of Requirement 64 of MIL-STD-454, shall be selected.

3.1.2.4 Mercury and radioactive material. The mercury and radioactive material requirements of MIL-E-16400 shall apply. In addition, no parts used shall have come in direct contact with metallic mercury, mercury compounds or mercury bearing instruments or devices employing only a single boundary of containment during manufacture or inspection. (A single boundary of containment is one which is not backed up by a second seal or barrier to prevent escape and consequent mercury contamination in the event of rupture of the primary seal or barrier.)

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3.1.2.5 Type BNC and type N radio frequency (RF) connectors. Where Type BNC and Type N RF coaxial connectors are specified herein, connectors shall be selected in accordance with MIL-C-39012.

3.1.3 Production equipment. Production equipments shall be identical to the equipment that has been subjected to and passed the first article examinations and tests (see 3.1.1 and 4.3).

3.2 Operational requirements. Each unit shall give reliable performance within the limits specified herein, under conditions of intermittent or continuous operation during exposure to the adverse conditions normally encountered in naval ship and shore communications service. Mechanical and electrical ruggedness is of paramount importance for assurance of reliable performance without failure.

3.3 Principal operating characteristics.

3.3.1 Frequency range. The frequency (tuning) range of the equipment shall be from 2.0 to 30.0 MHz. Each equipment shall be capable of adjustment to any frequency from 2.0 to 29.99999 MHz.

3.3.2 Tuning accuracy. The receiver shall tune to the desired operating frequency with an accuracy equivalent to that of the frequency standard.

3.3.3 Method of tuning. Frequency selection over the complete frequency range of the equipment shall be accomplished in increments of 10 Hz or less by front panel key pad entry and by a knob which shall provide manual tuning.

3.4 Primary power requirements. The equipment shall be designed to operate from 115 VAC, 60 Hz, single phase electrical power having characteristics as specified for Type I power in MIL-STD-1399, Section 300. The total power consumption shall not exceed 90 watts. A switch for disconnecting equipment from primary power shall be mounted on the equipment front panel and its function clearly labeled. It shall break all power conductors of the power circuit.

3.4.1 Transient protection. The equipment shall operate without damage or permanent change in characteristics due to transient voltage, transient frequency, or power interruption as specified in MIL-E-16400 and the spike tests of MIL-STD-1399, Section 300 using a 2000V spike for 440 volts and a 1000V spike for 115 volts. Protection shall be included against damage due to transients produced by operation of relays or tuning motors within the equipment.

3.4.2 Electrical overload protection. Electrical overload protection shall be provided in conformance with Requirement 8 of MIL-STD-454 for Class 1 equipment.

3.4.3 Leakage current. The equipment leakage current measured in accordance with the procedure of 4.5.4 shall not exceed 5 milliamperes (ma) AC or direct current (DC).

3.5 Modes of operation. The equipment shall be capable of receiving by selection of one of the following modes of operation:

a. LSB
b. USB
c. ISB
d. AM
e. CW telegraphy
f. FM.

3.6 Equipment arrangement. The equipment shall be capable of stacked installation in either of the following configurations without vertical space between units:

a. Multiple CY-4516 (A)/or equivalent hard-mounted installation in shipboard electrical equipment cabinets.

b. Mounting in an ANSI/EIA RS-310-C-77 mounting rack.

3.7 Radio receiver system. The contractor shall be responsible for demonstrating the full range of operations of the equipment to be delivered in accordance with the requirements of this specification. Equipment provisions for external cabling to the radio remote control system shall be required. Standard Navy cables shall be as specified in MIL-C-915. Mating connectors for all external cabling which requires manufacture by the installing activity shall be furnished with each equipment. Equipment design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of equipment.

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3.8 Dimensions and weight. The maximum dimensions and weight of the equipment shall not exceed those shown in Table I.

3.9 Environmental conditions. The equipment shall be capable of continuous, reliable operation, within the performance limits specified in 3.14 of this document and the following environmental service conditions of MIL-E-16400, paragraph 3.3.5:

a. Temperature range 4
b. Relative humidity
c. Salt fog (spray) for sheltered equipment
d. Fungus
e. Shock
f. Vibration
g. Magnetic field

The equipment shall operate within the specified tolerances throughout the full range of environmental conditions without alignment or adjustment, other than external controls normally employed for operation of the equipment.

3.10 Orientation. The equipment shall be capable of storage in any orientation.

3.11 Reliability.

3.11.1 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) (θ_0 as defined in MIL-STD-781) for this equipment shall be 5000 hours. The lower test, MTBF (θ_1), shall be 2500 hours.

3.12 Basic performance conditions. The basic test conditions specified in 4.6 shall apply to the performance requirements specified in 3.14, except as modified for a specific test or test procedure contained in this specification.

TABLE I. Dimensions and weight.

Unit	Width cm 1/ (in.) 2/	Depth 3/ cm (in.)	Height cm (in.)	Weight kg 4/ (lb) 5/
Receiver	48.3 (19.0)	49.5 (19.5)	15.24 (6.0)	18.2 (40.0)

1/ Centimeters
2/ Inches
3/ Exclusive of rear connectors
4/ Kilograms
5/ Pounds.

3.13 Mechanical design and construction. Mechanical design and construction furnished in accordance with this specification shall conform to MIL-E-16400 except as modified herein.

3.13.1 General requirements.

3.13.1.1 Construction. Construction shall conform to the requirements of MIL-E-16400. The equipment shall consist basically of a chassis, a front panel, and a group of replaceable plug-in assemblies. The chassis shall include the power supply, receptacles for the insertion of the plug-in assemblies, RF circuit protection, and the mechanical hardware associated with frequency selection. The front panel shall contain the required control functions and be securely mounted to the chassis. Two handles shall be provided, one mounted on each side of the front panel and each capable of supporting the entire weight of the unit. There shall be no sharp corners or edges on cases, panels, or brackets. Each printed circuit assembly shall be keyed to prevent incorrect insertion.

3.13.1.2 Rack mounting. The chassis of the equipment shall be designed and constructed so that when mounting brackets are installed on the sides of the chassis and the equipment is installed in a rack and subjected to shock or vibration, no other support or reinforcing members will be required to prevent damage to any part of the equipment.

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3.13.1.2.1 Cabinet mounting. The equipment shall also be designed to permit installation in a CY-4516(A)/U Electronic Equipment Cabinet as specified in MIL-C-24056.

3.13.1.2.2 Equipment enclosure. The degree of enclosure shall be dripproof (at 45 degrees) for internal ship spaces. The extraneous light requirement of MIL-E-16400 does not apply.

3.13.1.3 Bonding and grounding. Equipment bonding and grounding shall conform to the requirements of MIL-E-16400. Additionally, a stud or pad grounding facility shall be provided. The facility shall be a pressed-in captive insert integral with the chassis rear panel and constructed of austenitic corrosion-resistant steel. The insert shall be flush with the outside surface of the chassis. If a pad is used, it shall contain a blind threaded hole to accommodate a 5/16 inch - 18 bolt for ground strap fastening. If a stud is used, it shall be 5/16 inch - 18 threaded. A 25.4-millimeter (mm) (1 inch) diameter area centered on the blind hole or stud shall be free of paint and chemically treated to preserve the metal and provide a ground.

3.13.1.4 Corrosion protection, corrosion resisting treatment and painting. Corrosion protection and corrosion resisting treatments shall be in accordance with MIL-E-16400.

3.13.1.5 Fastener hardware. Fastener hardware shall conform to the requirements of MIL-E-16400. All screws, nuts, bolts, and other fastener hardware shall be of austenitic corrosion-resistant steel, except in those applications requiring material capable of withstanding high tensile stresses. Captive-type bolts shall be used on equipment front panels and electronic plug-in assemblies. Cadmium plating is not acceptable on the captive-type bolts used on the equipment front panels and electronic plug-in assemblies.

3.13.1.6 Electrical and RF connectors. Electrical and RF connectors shall conform to the requirements of MIL-E-16400. Electrical multi-pin connectors shall have at least 10 percent spare pins, with a minimum of one spare pin for each pin type.

3.13.1.6.1 RF input and output connectors. The coaxial RF connectors for the external frequency standard input and the antenna input shall be rear panel mounted and shall be Type TNC. A coaxial RF connector, Type BNC, shall be provided for the internal frequency standard output test.

3.13.1.6.2 Assembly (module) connectors. RF inserts, if used, shall be constrained from non-perpendicular movement to ensure proper mating.

3.13.1.7 Time meter. A time meter is not required.

3.13.1.8 Fuses. Fuses shall be in accordance with MIL-E-16400, except that extra fuses are not required. Fuseholders shall comply with MIL-F-19207.

3.13.1.9 Filters. Filters used in the sideband selection process shall be mechanical or crystal filter type.

3.13.1.10 Phone jacks. Phone jacks shall be provided on the front panel to permit the use of earphones in place of a speaker for full operation of the equipment.

3.13.1.11 Telephone plugs. Telephone plugs are not required.

3.13.1.12 RF meter. A RF meter shall be provided on the front panel to indicate relative RF levels.

3.13.1.13 Assembly identification. Assemblies shall be provided with the appropriate assigned National Stock Number, Manufacturer's Part Number (including revision indicator), and Federal Supply Code Number. This information shall be inscribed on the assemblies.

3.13.1.14 Dust covers. The dust covers of all assemblies employing printed circuit boards (PCBs) shall indicate the location of all test points and all adjustments accessible through the dust cover.

3.13.1.15 Accessibility. In addition to the requirements of MIL-E-16400, the circuit mounting boards and components shall be easily accessible for maintenance, repair, or replacement.

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3.13.1.16 Test points. Test points shall be so provided and located as to permit the isolation of malfunctioning parts or subassemblies within any major unit or assembly, without the unsoldering of any interconnection or wiring lead.

3.13.1.17 Circuit alignment. Resonant circuits shall be provided with an effective means of adjustment to permit accurate alignment. There shall be sufficient means of adjustment to compensate for the effects of normal aging of components and to facilitate normal preventive maintenance adjustments.

3.13.1.17.1 IF bandpass filters. Composite bandpass filters may be furnished as sealed units without trimmers.

3.13.1.17.1.1 IF output frequency. The IF output frequency shall be a nominal 455 kHz.

3.13.1.17.1.2 IF output impedance. The IF output impedance shall be 50 ohms unbalanced.

3.13.1.17.1.3 IF output level. The IF output level shall be not less than 20 mV for an RF input level at the AGC threshold.

3.13.1.17.1.4 IF output connector. The RF connector for the IF output test port shall be type BNC in accordance with MIL-C-39012.

3.13.1.18 Factory adjustment controls. The design of the equipment shall not include factory or sealed adjustment controls unless specifically approved by the procuring activity.

3.13.1.19 Shock and vibration.

3.13.1.19.1 Shock. The equipment shall be capable of withstanding the shock test for Grade A, Class I, lightweight, Type A equipment as specified in MIL-S-901.

3.13.1.19.2 Vibration. The equipment shall be capable of withstanding the Type I vibration test of MIL-STD-167-1. During vibration testing, the equipment shall be tested in accordance with the requirements of 4.4.5. The nameplate requirements of MIL-STD-167-1 are not applicable.

3.13.1.20 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400.

3.13.1.21 Welding. Welding shall conform to the requirements of MIL-E-16400.

3.13.1.22 Attachment of wires and leads. Attachment of wires and leads shall conform to the requirements of MIL-E-16400.

3.13.1.23 Internal wiring and cabling. Internal wiring and cabling shall conform to the provisions of MIL-E-16400 and Requirement 69 of MIL-STD-454. Hookup wire shall conform to MIL-W-16878, Type E.

3.13.1.24 Conductors. Conductors shall conform to the requirements of MIL-E-16400.

3.13.1.24.1 Conductor identification. Identification of conductors shall conform to the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the equipment.

3.13.1.25 Identification and marking. Identification and marking of all units, assemblies, electrical and moving mechanical parts, and mechanical parts which require removal for normal maintenance, shall be in accordance with MIL-E-16400 and MIL-E-21981. Hertz or Hz shall be used for all indications of frequency on hardware and associated documentation.

3.13.1.26 Part replacement. On board replacement parts (OBRPs) shall conform to the parts, materials and processes requirements of MIL-E-16400.

3.13.1.27 Quartz crystal units. Quartz crystal units shall conform to Requirement 38 of MIL-STD-454.

3.13.1.28 Gears and cams. Gears and cams shall conform to Requirement 48 of MIL-STD-454.

3.13.1.29 Maintenance accessories. Extender cables and/or extender cards, as applicable, shall be furnished as required. Electronic assemblies and printed circuit boards that plug in to chassis or

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mother board receptables and on which the test points or components are not available for fault isolation, require an extender cable or card for fault isolation.

3.13.1.29.1 PCB extender cards. The extender cards shall be of sufficient height to permit access to all test points and components on the PCB for dynamic fault isolation. The extender cards shall conform to the printed wiring requirements of MIL-E-16400.

3.13.1.30 Thermal design. Thermal design shall be in accordance with MIL-E-16400.

3.13.2 Human engineering. The contractor shall comply with the human engineering operational and maintenance design criteria of MIL-STD-1472 and the requirements specified herein except that equipment shall be designed for use by the 5th percentile female components to the 95th percentile male components.

3.14 Detailed performance.

3.14.1 Sensitivity and gain.

3.14.1.1 Sensitivity. The sensitivity of the equipment shall satisfy the requirements shown in Table II for any frequency within the complete frequency range of the equipment. Specified sensitivities shall be achieved in all modes with the manual RF gain control in the extreme clockwise (maximum gain) position.

3.14.1.2 Gain. The equipment gain shall be sufficient to produce standard audio output for the input voltages of Table II, for any frequency within the complete frequency range (2 to 30 MHz) of the equipment.

3.14.1.3 Gain variation. The variation in overall gain for any mode of reception shall not exceed 10 decibels (dB) for the complete .working frequency range of each MHz frequency band. Gain variations over the complete equipment tuning range (2 to 30 MHz) shall not exceed 15 dB total.

3.14.2 Selectivity.

3.14.2.1 Selectivity (antenna input to detector input). The selectivity of the equipment, excluding the AF response characteristics, shall depend primarily upon the characteristics of the filter defining the final IF passband. When measured in accordance with 4.5.8.2, selectivity shall be as specified in 3.14.2.1.1 and 3.14.2.1.2.

3.14.2.1.1 USB, LSB, ISB operation.

a. USB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz
b. LSB passband (-3 dB): Carrier -450 Hz to carrier -3000 Hz
c. ISB passband (-3 dB): Carrier +450 Hz to carrier +3000 Hz, and carrier -450 Hz to carrier -3000 Hz
d. Ripple (peak to valley): 3 dB maximum
e. Attenuation at carrier frequency: 10 dB minimum
f. Opposite sideband attenuation 1 kHz into opposite sideband: 60 dB minimum
g. Attenuation at carrier +5300 Hz (USB): 60 dB minimum
h. Attenuation at carrier -5300 Hz (LSB): 60 dB minimum.

3.14.2.1.2 AM and CW operation.

a. Passband (-3 dB): Carrier -3000 Hz to carrier +3000 Hz
b. Ripple (peak to valley): 3 dB maximum
c. Attenuation at carrier +9.5 kHz, carrier -9.5 kHz: 60 dB minimum.

TABLE II. Sensitivity.

MODE	CW	AM	USB, LSB, ISB	FM
IF BANDWIDTH (kHz)	0.3	6.0	3.0	16
SENSITIVITY (MICROVOLTS) (μV)	0.4	2.5	0.5	0.5
% MODULATION, 400 Hz MOD	N/A	50	N/A	N/A
DEVIATION (kHz), 400 Hz MOD	N/A	N/A	N/A	4.8
(S+N)/N (dB)	16	10	10	17

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3.14.2.2 Overall passband (antenna input to audio output). When measured in accordance with 4.5.8.2.1, the overall equipment passband characteristics shall be as specified in 3.14.2.2.1 and 3.14.2.2.2.

3.14.2.2.1 USB, LSB, and ISB characteristics. USB, LSB, and ISB equipment passband characteristics are as follows:

a. USB passband: Carrier +450 Hz to carrier +3000 Hz
b. LSB passband: Carrier -450 Hz to carrier -3000 Hz
c. ISB passband: Carrier +450 Hz to carrier +3000 Hz, and carrier -450 Hz to carrier -3000 Hz
d. Ripple (peak to valley):
   (1) Carrier +450 Hz to carrier +3000 Hz: 3 db maximum USB
   (2) Carrier -450 Hz to carrier -3000 Hz: 3 db maximum LSB.

3.14.2.2.2 AM and CW characteristics. AM and CW receiver passband characteristics are as follows:

a. Passband (-3 dB): Carrier -3000 Hz to carrier +3000 Hz
b. Ripple (peak to valley): 3 dB maximum.

3.14.2.3 Overload selectivity (DESENSITIZATION). Overload selectivity shall be such that when the equipment is operated in SSB modes, and with automatic gain control (AGC) on SLOW, an unmodulated out-of-band interfering signal shall not cause a decrease of more than 3 dB in the signal plus-noise-to-noise ((S+N)/N) ratio. The out of band signal shall be >50 kHz from the resonant frequency and >0 dBm.

3.14.2.4 Gain depression. At the desensitization points as specified in 3.14.2.3, gain depression shall not reduce the desired output by more than 8 dB.

3.14.3 Spurious responses. The equipment shall comply with the requirements of 3.14.3.1 through 3.14.3.6.

3.14.3.1 Internally generated spurious outputs. With no RF input signal, the antenna input terminated with a shielded 50 ohm resistor, and the equipment adjusted for the reception of SSB signal, internally generated equipment outputs shall conform to the requirements of 3.14.3.1.1 and 3.14.3.1.2.

3.14.3.1.1 Equipment operated on its internal frequency standard. No spurious responses shall occur at output levels that are greater than or equal to those which would result from a 2.0 μV (open circuit) reference signal applied to the antenna input.

3.14.3.1.2 Equipment operated on an externally supplied 1 MHz reference. When, in lieu of operating with the internal frequency standard, the equipment is supplied an external 1 MHz reference at an available power of +21 dB referenced to one milliwatt (5V open circuit in series with 50 ohms), having all non-harmonically related components at least 90 dB, and all harmonically related components at least 60 dB below the reference, internally generated spurious outputs shall conform to the requirements of 3.14.3.1.1, except that six spurious responses shall be permitted with output levels equivalent to a 2.0 μV (open circuit) reference signal applied to the antenna input.

3.14.3.2 Image rejection. Image rejection, measured at the audio output, shall be not less than 80 dB.

3.14.3.3 IF rejection. Rejection of all IFs, measured at the audio output, shall be not less than 80 dB.

3.14.3.4 Other single-signal spurious responses. Single-signal spurious responses, other than IF and image, shall be less than -114 dBM.

3.14.3.5 Front-end intermodulation (out of band). CW signals at two frequencies, F1 and F2, shall be applied simultaneously to the antenna input from a 50 ohm linear combiner. Sufficient filtering or isolation shall be used between each generator and the combiner to ensure adequate freedom from intermodulation products in the test signal. Signal pairs shall be (f_o +Δf, f_o +2 Δf), (f_o - Δf, f_o - 2 Δf), (f_o + 2 Δf, f_o + 3 Δf), and (f_o - 2 Δf, f_o - 3 Δf), with Δf equal to 5 percent f_o. A reference signal at the AGC threshold shall be applied at a frequency which results in a 1 kHz audio output, and the reference output amplitude recorded. The reference signal shall then be replaced by the signal pairs which shall be increased in amplitude, maintaining equality of the two signals, until the equipment output reaches the reference output amplitude, or until each input signal amplitude

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reaches -20 dBM. The equipment audio output shall not reach the reference for input signal amplitudes (each signal) of -20 dBM.

3.14.3.6 Cross modulation. Cross modulation shall be such that an off-resonant, 30 percent modulated signal spaced >50 kHz from the resonant frequency shall produce <10 percent cross modulation when the undesired signal is ≥ -20 dBM.

3.14.4 Audio outputs. The audio outputs from the USB and LSB amplifiers shall conform to the requirements of 3.14.4.1 through 3.14.4.4.2. Each amplifier shall feed line output receptacles at the rear of the equipment and its own head telephone jack on the front panel. The line outputs shall terminate in appropriately identified receptacles.

3.14.4.1 Audio output voltage regulation. Each audio amplifier shall be so designed that the output voltage shall remain constant within 30 percent (in V) of the value obtained with a 600 ohm load for a change in the value of this load from 100 to 600 ohms in 100 ohm steps. This condition shall be obtained with any input level from zero input to that which produces full output. The power output transistors shall not be subjected to damage or deterioration at any output load impedance from zero ohms to an open circuit.

3.14.4.2 Maximum undistorted output. With an RF input of not less than 1000 µV, the maximum undistorted output shall be not less than 1 milliwatt (mW), dissipated in a 600 ohm resistive load at the line output receptacles of the audio amplifier; or with the line level control set at the maximum rated output setting, the undistorted output appearing across a 600 ohm resistive load connected to the phone jack shall be not less than 30 mW. In the AM mode, with 30 percent modulation, distortion shall be no greater than 5 percent.

3.14.4.3 Audio level controls. Separate (non-coaxial) level controls shall be provided on the equipment front panel for audio line and local (phone) outputs. Phone level controls shall be equipped with knobs chosen in accordance with MIL-E-16400 part selection procedures. Line level controls shall be provided on the front panel for audio line controls and may be either knob or screwdriver-adjustable types. Variable resistors shall comply with the requirements of MIL-R-94. Controls shall be standard audio (clockwise log) taper, and shall provide smooth variation in output.

3.14.4.3.1 Audio level metering. The equipment shall be provided with a front panel audio metering system which will indicate relative audio line output levels.

3.14.4.4 Output hum and noise.

3.14.4.4.1 Output signal-to-hum ratio. The output signal-to-hum ratio for AM, USB, or LSB shall be not less than 45 dB for the standard audio output of 1 mW. This requirement shall apply separately to each output component which is related to the power line frequency, including hum sidebands on the desired audio output. Measurement shall be in accordance with 4.5.8.4.5.

3.14.4.4.2 Output signal to hum-plus-noise ratio. The output signal to hum-plus-noise ratio for AM, USB, and LSB modes, measured in accordance with 4.5.8.4.5, shall be not less than 35 dB.

3.14.5 Manual RF/IF gain control. The manual RF/IF gain control shall have the range to provide a variation in the overall gain of the RF and IF amplifiers of the equipment of not less than 100 dB. The RF gain control shall be accessible from the front panel and shall be disabled in the AGC mode.

3.14.6 AGC characteristics. The AGC characteristics shall be as stated in a and b below.

a. Audio output shall remain constant within a total tolerance of 3 dB for an increase of 100 dB above AGC threshold.

b. The total harmonic distortion shall increase not more than 5 percent at RF input levels up to 100,000 µV, open circuit.

3.14.6.1 AGC conditions. The requirement of 3.14.6 shall apply for conditions a through d, singly or in any combination:

a. Any frequency selected within the nominal frequency range of the equipment.

b. Any audio gain adjustment producing audio levels at or below maximum undistorted output.

c. Any constant level of carrier modulation between 30 and 90 percent.

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d. Any constant modulation frequency between the limits of 450 and 3000 Hz.

3.14.6.2 AGC time constants. The AGC time constant shall be as specified in Table III. These time constants shall hold for signal transitions of 60 dB or less.

TABLE III. AGC time constants.

AGC	ATTACK TIME	DECAY TIME
FAST	<20 msec	<35 msec
SLOW	<20 msec	5 SECONDS MAXIMUM

3.14.6.3 AGC linearity. There shall be no evidence of blocking or amplifier saturation at any input level up to 100 dB above AGC threshold.

3.14.6.4 Resonant overload. Audio output power (remote outputs) at resonant overload (see 6.4.10) shall be a minimum of 1 mW.

3.14.6.5 Time constant of recovery from surge inputs. The design shall be such that with the equipment operating under standard conditions as for the reception of CW signals and with AGC off, a sharp pulse of CW input signal, at resonance, having a minimum amplitude of 5V and duration of 50 ms (to simulate static surge), will not cause a reduction in sensitivity that will persist in excess of 40 ms after cessation of the input surge.

3.14.6.5.1 Measurement of time constant. The time constant shall be measured from the time of removal of the input signal pulse until the output noise level has reached 63 percent of the steady-state value. Start of time shall begin at the moment of removal of the input signal pulse.

3.14.7 BFO control. The BFO control shall operate (when the mode is set at CW) so as to permit a variation of the frequency of the BFO of not less than ±6000 Hz.

3.14.7.1 BFO frequency variation. The variation of frequency shall be made by key pad entry or manual tuning. The frequency of the BFO shall increase in the clockwise direction and decrease in the counterclockwise direction of the control, when using the tuning control.

3.14.7.2 BFO buffering. The BFO shall be buffered to prevent pulling of the BFO frequency by more than 10 Hz due to equipment RF input signal amplitudes of 10,000 µV or less.

3.14.8 In-band intermodulation. Intermodulation distortion products generated when the equipment processes two equal-amplitude unmodulated RF signals producing audio outputs at 935 and 1045 Hz shall each be a minimum of 40 dB below the reference output tones. This requirement shall be satisfied in USB and LSB operation with the AGC time constant selector switch in both FAST and SLOW positions, over an input amplitude range of 100 to 100,000 µV open circuit, per tone, and for audio output levels of 0 dBm or less, per tone.

3.14.9 RF input circuit. The equipment antenna input circuits shall be designed for maximum sensitivity to signals from a 50 ohm impedance. The input circuits shall be protected from RF voltages up to 100 V rms, at the antenna input connector of the equipment. The overload protection circuit shall automatically reset when the overload voltage no longer exists. At least the above degree of protection shall exist with the equipment on or off. Fuses are not acceptable for RF overload protection. Operation of the protection circuit by an out-of-band transmitter shall not produce more than 40 dB desensitization of any equipment operating frequency; measurement shall be in accordance with 4.5.8.9. Equipment sensitivity shall not degrade by more than 1 dB relative to reference measurements made prior to testing in accordance with 4.5.8.9.

3.14.10 Accuracy of main tuning indicator calibration. The accuracy of frequency indications shall be within the limits specified in 3.14.10.1. The specified accuracies shall apply to all tuning indicators, acting in combination, to indicate the tuning of the equipment.

3.14.10.1 Tuning steps. Tuning in steps of 10 Hz or less shall be provided with an accuracy equivalent to that of the internal frequency standard. It shall be possible to tune the equipment to any frequency within the tuning range in less than 20 seconds.

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3.14.11 Frequency and gain stability. The equipment shall include provisions for operation from either an internal or external frequency standard of 1 MHz. All frequency stability requirements apply to operation using the internal standard.

3.14.11.1 Frequency stability and gain variations. The requirements of 3.14.11.1 through 3.14.11.1.6.2 shall apply under conditions of CW or sideband reception, as specified. The frequency stability figures shall apply to the variation in a 1000 Hz AF beat note with a constant input frequency. The frequency changes shall be made with reference to the frequency variations of all oscillators on the audio beat note frequency. All measurements shall refer to total variations, either instantaneous, transient, or gradual changes. The requirements specified herein shall apply at any frequency within the frequency range of the equipment.

3.14.11.1.1 Sixty-minute warm-up period. Measurements shall be made in the sideband mode at constant ambient temperature, constant relative humidity and constant line input voltage. Two minutes after turn-on time, the equipment shall be within 0.1 percent of the frequency it will have reached at the end of 60 minutes. Fifteen minutes after turn-on time, the equipment shall be within 0.002 percent of the frequency reached 60 minutes after turn-on. Sixty minutes after turn-on, the equipment shall be within one part in 10^7 of the desired frequency.

3.14.11.1.2 Variation in primary power voltage and frequency. Variation of primary power supply line voltage over a range of 115V, ±10 percent and of power line frequency over a range of 58 to 62 Hz, in any combination, shall not cause a change in the frequency stability of the equipment of more than two parts in 10^8 (total), or a change in overall gain (AGC defeated) in excess of 2 dB.

3.14.11.1.3 Variation in relative humidity. Exposure of the equipment to humidity conditions cited in 3.9 shall not cause loss of synthesizer frequency lock or excessive changes in frequency or gain.

3.14.11.1.4 Variation in ambient temperature. The equipment, when in the operating condition and when subjected to the operating temperature extremes for Range 4 equipment specified in MIL-E-16400, shall not exhibit frequency or gain variation in excess of the limits shown in Table IV. The specified limits for variations of frequency and gain shall be the total deviations of these parameters, which occur during the time period required by the high and low operating temperature cycling test procedures specified in MIL-E-16400. The AGC shall be off during gain variation measurements.

TABLE IV. Frequency and gain variation.

Performance requirement	Variation
Frequency variation versus change in ambient temperature from 0° to +50°C	1 part in 10^7
Gain variation versus change in ambient temperature from 0° to +50°C	8 dB (Total)

3.14.11.1.5 Control variations. Frequency variations resulting from a variation of any non-frequency-determining control throughout the full range of operation shall not exceed five parts in 10^8. This requirement shall apply to those controls which operate in non-frequency-determining circuits, and to controls which affect the RF or AF gain or level. The specified limit for frequency variation shall be the absolute maximum deviation of this parameter which occurs during the manipulation of any control throughout its full range of operation.

3.14.11.1.6 Short and long term stability. The equipment shall be capable of operation for a period of 24 hours without operator attention. "Without operator attention" shall be interpreted as meaning that reliable reception of a desired signal shall be assured throughout a 24-hour period without the necessity for readjustment of any tuning control. This requirement shall be applicable throughout the complete frequency range of the equipment.

3.14.11.1.6.1 Sudden changes during 24-hour period. During the 24-hour period of operation specified in 3.14.11.1.6, the equipment shall not exhibit sudden sporadic shifts in the frequency generated by any of the oscillators.

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3.14.11.1.6.2 Frequency drift rate. After an initial stabilization period, not to exceed 48 hours following turn-on, the overall frequency drift rate shall not exceed 5 parts in 10^8 (total) per day. This rate shall be achieved throughout the equipment operating frequency range, in any mode of operation which does not use the BFO or frequency vernier. The long-term frequency drift rate, averaged over a period of 60 days, shall not exceed 3 parts in 10^9 (total) per day. During measurement of frequency drift rate, changes in power supply line voltage and frequency shall be limited to ±1 percent, and temperature variations shall be limited to 10°C (total). Under the same conditions, equipment gain (AGC off) shall not change by more than 1 dB (total) during a 24 hour period after initial stabilization.

3.14.12 Regeneration. There shall be no evidence of unintended oscillations or instability under any condition of equipment operation, including primary line voltage 10 percent higher than the nominal primary line voltage.

3.14.13 Oscillator radiation. Oscillator radiation shall conform to the Class A4 requirements of MIL-STD-461. Antenna terminal narrowband emissions in the region 10 kHz to 1 GHz shall not exceed -90 dBm. Broadband emissions in the same frequency range shall comply with the requirements of MIL-STD-461, Method CE06.

3.14.14 Frequency standard adjustment range. The internal 1 MHz frequency standard shall be capable of ±20 Hz adjustment as a minimum. The adjustment shall be accessible from the top surface of the frequency standard assembly without removing the assembly cover. Piston type adjustments shall be protected from damage at the ends of their adjustment ranges.

3.14.14.1 Operation using an external frequency standard. The equipment shall be capable of operation from an external frequency standard of 1 MHz with any input level between 400 mV and 10.0 V, open circuit, in series with 50 ohms. Input impedance at the external 1 MHz input terminals shall be 50 ohms, nominal, within a voltage standing wave ratio (VSWR) of 2.0:1 or less.

3.14.14.2 Reference output signal. The equipment shall provide a 1 MHz reference output signal of between +7 and +13 dBm, delivered to a 50 ohm load, for testing.

3.14.14.3 Frequency standard switch functions. The internal frequency standard assembly shall have a function switch accessible without removal of the assembly dust cover. This switch shall have two positions with the following functions:

a. EXT: permits operation from an external frequency source. The internal frequency standard oscillator and oven shall be kept on for immediate availability with full accuracy.

b. INT: permits operation from the internal standard.

3.14.15 Frequency generation. The frequency generation process shall utilize only a single reference frequency, with all internally generated frequencies used in the synthesis of the operating frequency derived by arithmetic operations on, and phase-locked to the reference.

3.14.16 Electromagnetic compatibility (EMC). The equipment shall conform to the requirement of MIL-STD-461 for Class A4, Communications-Electronics equipment.

3.14.17 Remote operation. The receiver shall be capable of remote control operation. The following functions shall be controlled by the remote unit:

a. IF Bandwidth
b. Mode selection (AM, FM, CW, USB, LSB, ISB)
c. AGC (FAST-SLOW) or Manual gain control
d. Tune frequency
e. BFO control
f. Scanning (Stop/Start) (see 3.14.18)
g. Number of channels to be scanned (see 3.14.18)
h. Signal threshold for each channel (see 3.14.18).

3.14.17.1 Remote electrical interface. The electrical interface for remote operation shall conform to MIL-STD-188.

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3.14.18 Scan functions. The scan mode of the equipment shall have the following characteristics:

a. The equipment shall be capable of scanning from 2 to 99 frequencies in the frequency range specified in 3.3.1.

b. The equipment shall have a 99 channel memory. The receiver shall be capable of retaining stored information a minimum of 0.5 hour after loss of primary power.

c. The equipment shall be able to scan up to a rate of 5 channels per second. The dwell time shall be adjustable from 0.2 to 10 seconds per channel.

d. When scanning, receipt of a signal whose strength is above that of the threshold for that channel shall cause the receiver to stop scanning and lock on that signal. Threshold shall be adjustable from AGC threshold to 80 dB above the AGC threshold for each channel.

3.14.19 Carrier operated relay (COR). Means shall be provided to permit the operator to select a relative signal strength for operation of a relay. The relay contacts shall be brought out at the rear panel.

3.15 Maintainability.

3.15.1 Quantitative maintainability requirement. The equipment shall have a Mean-Time-To-Repair time (MTTR) not exceeding 0.5 hour and a maximum corrective maintenance time (M_{max} ct) not exceeding 1.5 hours when corrective maintenance is accomplished at the organizational level of maintenance by the replacement of lowest subassemblies (modules, PCBs) and chassis-mounted parts (electronic, electrical, electromechanical, and mechanical parts). The Mean-Time-To-Repair (MTTR) includes localization, isolation, disassembly, interchange, reassembly, alignment, and checkout for all corrective maintenance actions. The equipment is in a corrective maintenance mode whenever a failure, including a malfunction that allows degraded operation, occurs.

3.15.2 Built-in test (BIT) feature. The equipment shall contain a BIT feature which is capable of fault detection and isolation at the level of the lowest replacement module and reports operational status with both local and remote display.

3.16 Workmanship.

3.16.1 Workmanship. The workmanship and soldering requirements shall be in accordance with Requirements 5 and 9 of MIL-STD-454. Printed circuit assemblies shall comply with MIL-I-46058 and MIL-P-28809.

3.16.2 Workmanship screen. All receivers produced in accordance with this specification shall withstand a defect detection vibration screen of random type vibration at $0.04g^2/Hz$ ± 3 dB from 80 to 350 Hz and temperature cycling as specified in 4.5.3.2.

3.17 Embedded computer (EC) resources. The use of embedded computer resources as defined in TADSTAND A is a design consideration and not a specific requirement of this specification. If the contractor's design contains embedded computer resources, the requirements of this paragraph shall apply. Input/output interfaces shall comply with the provisions of TADSTAND B; software/firmware shall conform to applicable portions of MIL-STD-1679 (Navy).

3.17.1 Embedded computer selection. Use of any non-Navy standard embedded computer as defined in TADSTAND B requires prior Government approval.

3.17.2 Computer programming language. Use of any programming language other than a Navy standard high order language as defined in TADSTAND C requires prior Government approval.

3.17.3 Reserve capacities. As prescribed in TADSTAND D, use of embedded computer resources shall require the reserve capacities stated in 3.17.3.1 through 3.1.7.3.4 below.

3.17.3.1 Main memory. Main memory shall have a 20 percent reserve capacity, as a minimum. The reserve capacity shall be measured at peak main memory loading of the EC during its operational missions. Peak main memory loading shall include all programs and data required for successful operational mission execution. The reserve capacity need not be physically installed in the EC, but no cabinet, chassis, or backplane modifications shall be required for its installation.

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3.17.3.2 Throughput. Central processor throughput shall have a 20 percent reserve capacity, as a minimum. The reserve capacity shall be expressed as a percentage of available capacity at full operational loading over a specific period of time, as determined by operational mission characteristics. In the case of cyclic applications, such as some signal processing applications, this period would be the time between availability of successive input data buffers.

3.17.3.3 Number of input/output channels. The number of reserve input/output channels shall be 18.75 percent (3/16) of those available, as a minimum. The reserve channels need not be physically installed in the EC, but no cabinet, chassis, or backplane modifications shall be required for their installation.

3.17.3.4 Input/output channel throughput. Input/output channels shall each have a 20 percent reserve capacity in throughput, as a minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein and, unless disapproved by the Government, the contractor may use his own or any other facilities suitable for the performance of these inspection requirements. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that materials and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to:

a. Surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied,
b. Government product inspection to measure quality of the product to be offered for acceptance, and
c. Government inspection of delivered products to assure compliance with all inspection requirements of this specification.

Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification shall be as defined in MIL-STD-109.

4.2 Inspections and tests. Inspections and tests specified in Table V shall be conducted in accordance with the applicable test methods of MIL-E-16400 and MIL-STD-415 except as otherwise specified herein.

4.2.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection
   (1) Production inspection (Group A) (see 4.4.1)
   (2) Production control inspection (Group B) (see 4.4.2)
   (3) Environmental inspection (Group C) (see 4.4.3)
c. Inspection of packaging (see 4.9).

4.3 First article inspection. First article inspection shall consist of all examinations and testing necessary to determine compliance with all applicable requirements of this specification, including the tests specified in Table V (see 6.2).

4.4 Quality conformance inspection.

4.4.1 Production inspection (Group A). Production inspection shall be made on every item offered for delivery. This inspection shall encompass such examinations and tests which will prove the workmanship and reveal any omissions or errors of the production process such as functional and performance tests at a limited number of points in the required range, tests which detect deviations

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TABLE V. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Weight	3.8	4.5.1	X	X		
Size	3.8	4.5.1	X	X		
Parts and materials	3.1.2	4.5.1	X	X		
Marking	3.13.1.25	4.5.1	X	X		
Weld	3.13.1.21	4.5.1	X	X		
Mechanical design	3.13	4.5.1	X	X		
Bonding and grounding	3.13.1.3	4.5.1	X		X	
Corrosion protection	3.13.1.4	4.5.1	X		X	
Painting	3.13.1.4	4.5.1	X		X	
Assembly identification	3.13.1.13	4.5.8.22	X		X	
Soldering	3.16.1	4.5.1	X		X	
Conductor identification	3.13.1.24.1	4.5.1	X		X	
Frequency range	3.3	4.5.8.10	X	X		
Overload protection	3.4.2	4.5.1	X		X	
Leakage current	3.4.3	4.5.4	X		X	
Sensitivity	3.14.1.1	4.5.8.1	X	X		
Gain	3.14.1.2	4.5.8.1.1	X		X	
Gain variation	3.14.1.3	4.5.8.1.2	X	X		
Selectivity	3.14.2	4.5.8.2	X	X		
Overall passband	3.14.2.2	4.5.8.2.1	X	X		
Desensitization	3.14.2.3	4.5.8.2.2	X		X	
Gain depression	3.14.2.4	4.5.8.2.3	X		X	
Spurious responses	3.14.3	4.5.8.3	X		X	
Internal spurious responses	3.14.3.1	4.5.8.3.1	X	X		
Image rejection	3.14.3.2	4.5.8.3	X	X		
IF rejection	3.14.3.3	4.5.8.3	X	X		
Single signal responses	3.14.3.4	4.5.8.3	X	X		
Front end IM	3.14.3.5	4.5.8.3	X		X	
Cross modulation	3.14.3.6	4.5.8.3.1	X		X	
Audio output regulation	3.14.4.1	4.5.8.4.1	X		X	
Maximum undistorted output	3.14.4.2	4.5.8.4.2	X	X		
Audio level controls	3.14.4.3	4.5.8.4.3	X		X	
Audio level meter	3.14.4.3.1	4.5.8.4.4	X		X	
Signal-to-hum	3.14.4.4.1	4.5.8.4.5	X	X		
Signal-to-hum and noise	3.14.4.4.2	4.5.8.4.5	X		X	
Manual RF gain control	3.14.5	4.5.8.5	X		X	
AGC characteristics	3.14.6	4.5.8.6	X	X		
AGC time constants	3.14.6.2	4.5.8.6.1	X		X	
Scan	3.14.18	4.6.8	X	X		
COR	3.14.19	4.5.7	X	X		
Resonant overload	3.14.6.4	4.5.8.6.2	X		X	
Surge recovery	3.14.6.5	4.5.8.6.3		X	X	
BFO control	3.14.7	4.5.8.7	X	X		
In-band IM	3.14.8	4.5.8.8	X		X	
RF input circuit	3.14.9	4.5.8.9	X		X	
Accuracy of tuning	3.14.10	4.5.8.10	X		X	
Tuning time	3.14.10.1	4.5.8.10.1	X		X	
Frequency stability	3.14.11	4.5.8.11	X			X
Gain stability	3.14.11	4.5.8.11	X			X
Sixty-minute warm-up	3.14.11.1.1	4.5.8.11.1	X		X	
Line voltage variation	3.14.11.1.2	4.5.8.11.2	X			
Line frequency variation	3.14.11.1.2	4.5.8.11.2	X			
Variation in humidity	3.14.11.1.3	4.5.8.11.3	X			
Variation in temperature	3.14.11.1.4	4.5.8.11.4	X			
Control variation	3.14.11.1.5	4.5.8.11.5	X		X	
Frequency drift rate	3.14.11.1.6	4.5.8.11.3	X			X
Regeneration	3.14.12	4.5.8.12	X		X	

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TABLE V. Examinations and tests - Continued.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Oscillator radiation	3.14.13	4.5.8.13	X		X	
External frequency standard	3.14.14.1	4.5.8.14.1	X		X	
Frequency standard	3.14.14.3	4.5.7	X		X	
Steady-state voltage switch	3.4	4.5.8.18	X		X	
Power and power factor	3.4	4.5.8.20	X		X	
Steady-state frequency	3.4.1	4.5.8.18	X		X	
Transient voltage	3.4.1	4.5.8.18	X		X	
Transient frequency	3.4.1	4.5.8.18	X		X	
Power interruption	3.4.1	4.5.8.19	X		X	
Temperature sequence	3.9	4.5.3.2	X			X
Low temperature	3.9	4.5.8.16	X			X
High temperature	3.9	4.5.8.16	X			X
Humidity	3.9	4.5.8.16	X			X
Salt fog (spray)	3.9	4.5.1	X			X
Reliability	3.11	4.7	X			
Performance conditions	3.12	4.6	X			
Enclosure	3.13.1.2.2	4.5.8.17	X			X
Shock	3.13.1.19.1	4.5.8.17	X		X	
Vibration	3.13.1.19.2	4.5.8.17	X		X	
Thermal design	3.13.1.30	4.5.3.2	X			X
EMC	3.14.16	4.5.8.23	X			X
Remote operation	3.14.17	4.5.8.24	X		X	
Scan functions	3.14.18	4.5.8.25	X		X	
Maintainability	3.15	4.8	X		X	
Workmanship	3.16	4.5.2	X	X		
Workmanship screen	3.16.2	4.5.3	X	X		

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from design, tests of controls or adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examination and tests shown in Group A of Table V, subject to the conditions of 4.4.1.1.

4.4.1.1. Group A tests. The Group A tests shall be performed in accordance with the test methods of 4.5.8 except as modified by the following:

a. Image rejection: High frequency mixer image only
b. IF rejection: Of the first IF only
c. Maximum undistorted output: 2.1 MHz
d. Output signal-to-hum ratio: 2.1 MHz
e. AGC characteristics: 2.1 MHz
f. Frequency stability: 5 MHz.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in MIL-STD-105. This inspection shall consist of the examinations and tests which encompass functional and performance tests throughout the entire range of operation; tests which will detect any deterioration of the design by wear of such items as dies, molds, and jigs, and by substitution of different parts; tests which detect deviations in the processing of the materials; and tests to determine temperature rise produced in operation and ability of the receiver to withstand this heat. These tests shall be performed on the complete equipment as offered for delivery. Production control inspections shall be conducted on inspection lots that have passed the production inspection of 4.4.1.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defects or screen out the defective units and re-inspect the lot using a more stringent inspection. Rejected lots shall be kept separate from the new lots and shall not be allowed to lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be conducted in accordance with MIL-STD-810 on a sampling basis as specified in 4.4.3.1. This inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the equipment as a whole; life tests; simulated service tests; tests of the effects of changes in environment (such as extremes of temperature and humidity or effect of salt air), and tests of the effects of shock and vibration. Environmental inspection shall include the examination and tests shown in Group C of Table V.

4.4.3.1 Sampling for inspection of equipment. The equipment selected for environmental inspection must pass all tests, except EMI, before the lot of equipments from which the sample is selected can be moved from the contractor's facilities. One equipment from the first month's production lot, or from the first 50 units produced, shall be subjected to environmental inspection. One equipment from each successive production lot of 100, or one per month if less than 100 are produced monthly, shall be subjected to environmental inspection. One equipment from each 50 equipments produced shall be subjected to EMI tests. Environmental inspection shall be performed on equipments that have passed the production inspection of 4.4.1.

4.4.3.2 Sample failure. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative (QAR) the results thereof and corrective action taken to correct units of product which were manufactured with the same conditions, materials, and processes and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental inspection sample units. Unless otherwise specified, sample units which have been subjected to and passed production control or environmental inspections or both, may be accepted on the contract provided that they are resubjected to, and pass production inspection after repair of all defects.

4.4.5 Vibration test conditions. During vibration tests, the equipment shall be operated in the USB mode with the AGC time constant selector in the SLOW position, and with an unmodulated RF input signal producing an audio output of 1000 to 3000 Hz. The RF input signal shall be continuously monitored with an oscilloscope and a wave analyzer. Any vibration-related product present at the audio output shall be not less than 30 dB below the reference audio output amplitude. No signal loss (dropout) exceeding 3 dB may occur during the vibration test.

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4.5 Test methods.

4.5.1 Test and examinations. Where 4.5.1 is specified in Table V, the test method specified in MIL-E-16400 shall be applied.

4.5.2 Workmanship.

4.5.2.1 General workmanship and soldering. The equipment, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process. Each solder connection and its associated wiring leads shall be visually examined.

4.5.3 Workmanship screen.

4.5.3.1 Random vibration. Prior to conducting temperature cycling, vibration shall be performed on each equipment. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random, or subject to procuring activity approval, pseudo-random or complex wave-form vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. All items shall be hard-mounted (without shock insulators) and subjected to the vibration conditions of Figure 1. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the acceleration level. The test item shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

4.5.3.2 Temperature cycling. Each unit shall be subjected to 10 cycles of the temperature curve indicated in Figure 2. The temperature rate of change shall be not less than 5°C per minute. Unit power shall be turned on and off at the indicated times. The unit shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the unit reaches 80 percent of the chamber temperature. When failures occur, the unit shall be reworked and the cycling continued for a cumulative total of 10 cycles. Equipment operating time of 20 hours shall be accumulated during these 10 cycles.

4.5.4 Leakage current test. Equipment shall be labeled as follows:

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES.

THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR AN INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.4.1 Equipment connections. Each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipment and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment under test and the source power shall be opened during the test. The equipment shall be connected as shown in Figure 3.

4.5.4.2 Measurement. Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500 ohm

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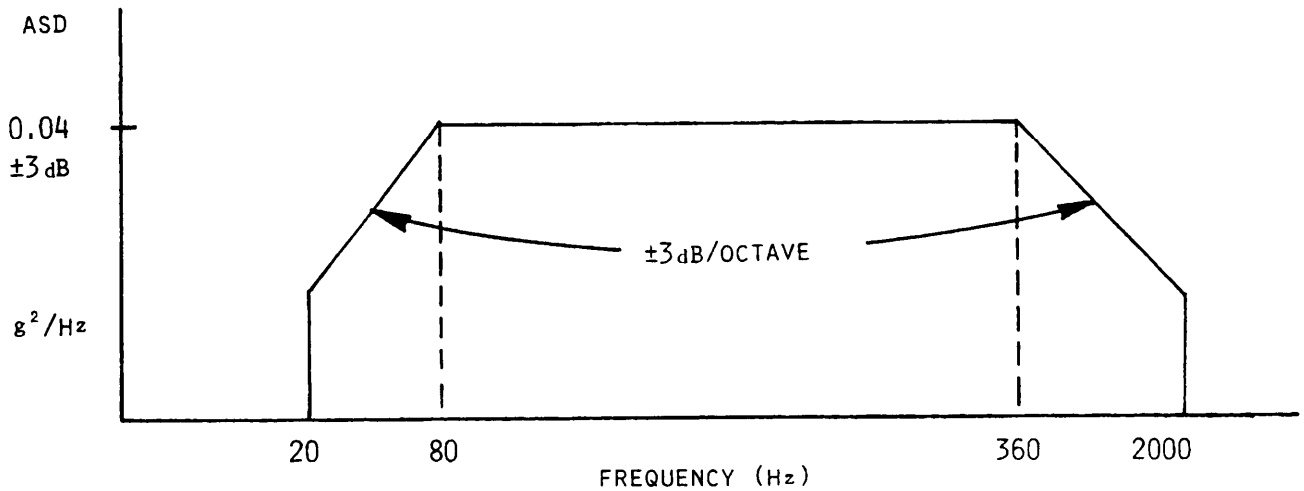


FIGURE 1. Random vibration curve

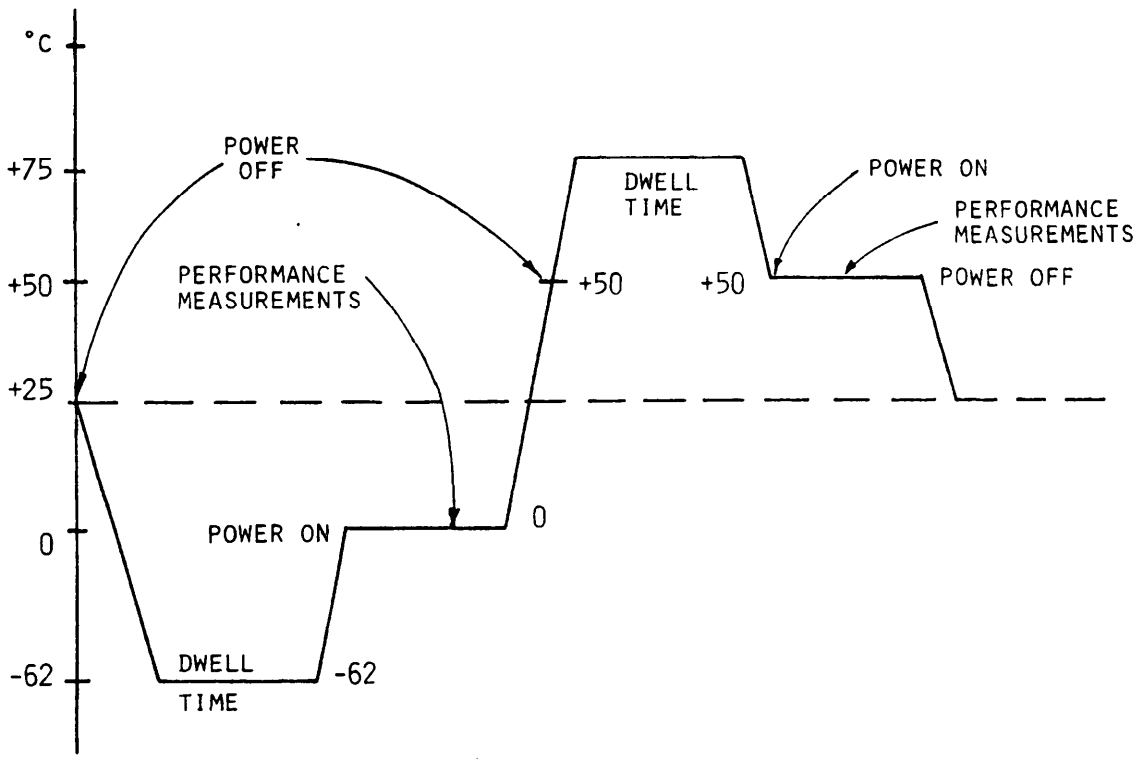


FIGURE 2. Temperature curve (one cycle) (not to scale)

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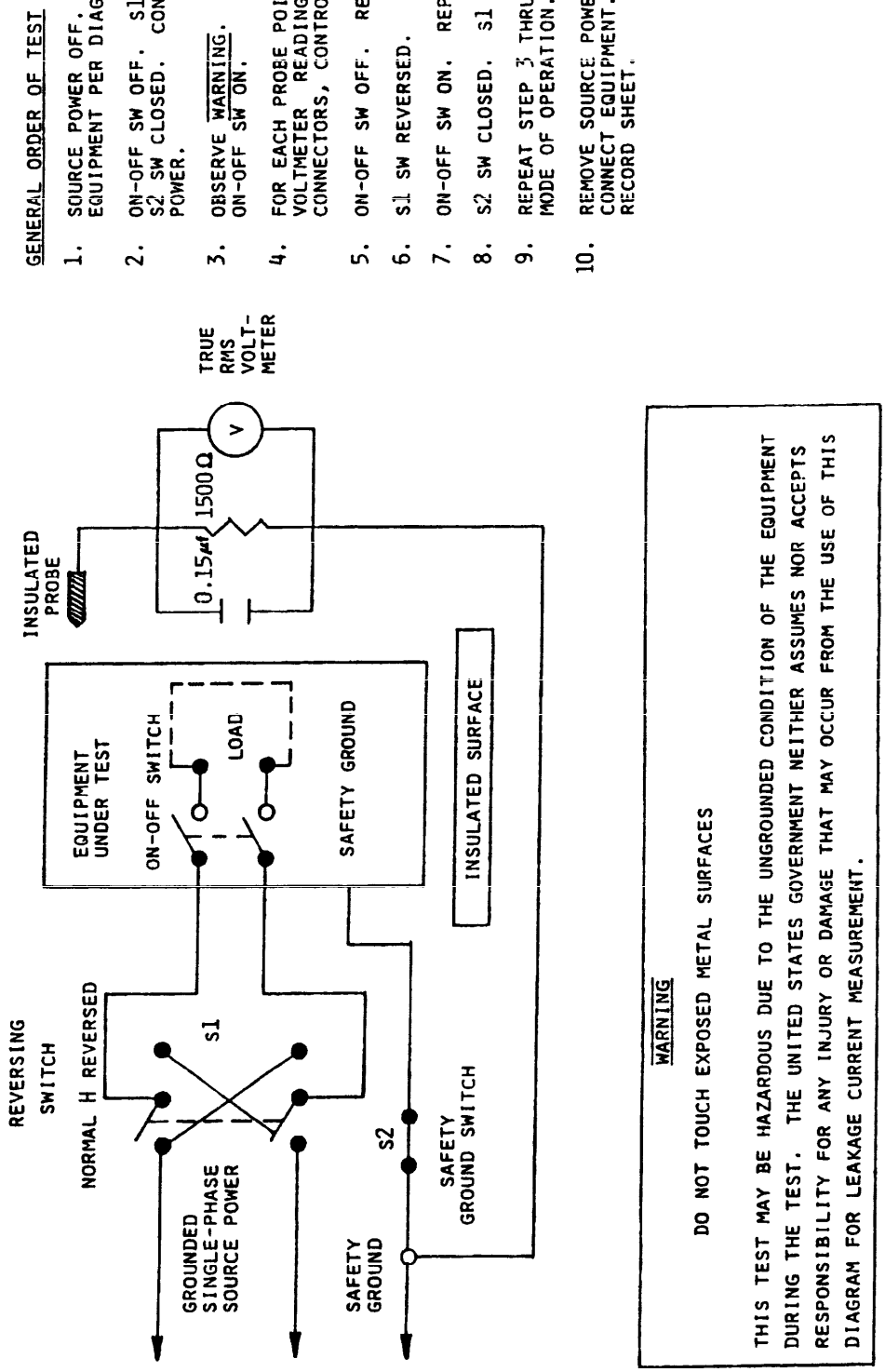


FIGURE 3. Single-phase test diagram for leakage current measurement

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resistor shall not exceed 7.5 volts at the highest nominal power line voltage and the highest and lowest nominal power line frequencies for which the equipment is designed. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed and the voltage measured for every possible combination of switch positions shown in Figure 3. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.5 Cables. Interconnecting cables to be supplied with the end item shall be installed on the end item prior to any testing, and shall be used for the duration of all testing.

4.5.6 Contractor validation. The contractor shall inspect the equipment to determine technical compliance and adequacy for the specified characteristic or requirement. This validation shall include qualitative examination and testing whenever a function is specified.

4.5.7 Inspection by testing. When this paragraph is specified in Table V, the contractor shall perform qualitative testing.

4.5.8 Detailed test methods.

4.5.8.1 Sensitivity. Measurements shall be made at the 5, 50, and 95 percent points in each of the primary frequency bands for the USB mode, and at the frequency of poorest USB sensitivity for LSB, AM and CW modes.

4.5.8.1.1 Gain. Measurement shall be made at the frequency of lowest gain as determined by the procedure of 4.5.8.1.2.1.

4.5.8.1.2 Gain variation. Measurements shall be made using the procedure of 4.5.8.1.2.1 to determine the gain variation of each MHz frequency band. Measurements shall be made with the equipment in USB mode, with AGC off.

4.5.8.1.2.1 Gain variation measurement procedure. Apply a signal of one mV, open circuit, in series with 50 ohms, at a frequency of 2.1010 MHz through the equipment antenna input terminal. Tune the equipment to 2.1000 MHz in USB mode. Adjust the RF gain control for maximum and turn AGC to OFF. Adjust the audio line level control to produce 1 mW of audio output. Leaving the gain and level controls and the input signal amplitude fixed, tune the equipment and signal source together to produce a constant output frequency of 1 kHz. Within each MHz frequency band, each setting of the 100 kHz shall be tested. Repeat the procedure, determining the minimum output in each band.

4.5.8.2 Selectivity (antenna input to detector input). Measurements shall be made in USB, LSB, and AM modes with the equipment tuned to 17.5500 MHz. AGC shall be inoperative and RF gain shall be maximum. Measurements shall be made of all parameters specified in 3.14.2.1.1 and 3.14.2.1.2, using an unmodulated test signal. Ripple in the passband shall be measured between the -3 dB points using a constant amplitude RF input signal which produces an output (S+N)/N ratio of 20 dB at the peak of the passband ripple. The amplitude and frequency (relative to the carrier) of each inflection point shall be determined. Measurements of carrier and opposite sideband rejection and 60 dB attenuation points shall be made by increasing the RF input signal amplitude to obtain the reference IF output produced at the peak of passband ripple.

4.5.8.2.1 Overall passband (antenna input to audio output). Measurements shall be made in USB and LSB modes, with the equipment tuned to 17.5500 MHz. AGC shall be inoperative. Bandwidth (-3 dB) and passband ripple shall be measured using a constant amplitude unmodulated RF input signal which produces an output (S+N)/N ratio of 20 dB at the peak of the passband ripple. Passband ripple measurements shall be made in frequency increments not larger than 100 Hz.

4.5.8.2.2 Overload selectivity (desensitization). Overload selectivity (desensitization) shall be measured with the equipment operating in the SSB mode, and with AGC on SLOW. An unmodulated signal shall be applied at resonance to the input of the equipment from a 50 ohm standard dummy antenna and adjusted to yield standard sensitivity. An unmodulated interfering signal shall be applied from the same standard dummy antenna at a frequency differing from the original signal by the IF bandwidth; the signal level shall be increased to the point where the (S+N)/N ratio of the original signal is decreased by not more than 3 dB. The procedure shall be repeated for increasing frequency spacing of the desensitizing signal from the equipment operating frequency, including a minimum number of points specified in 3.14.2.3.

4.5.8.2.3 Gain depression. At each point on the overload selectivity curve, the amount of gain depression for the desired signal shall be measured at the 3 dB desensitization point by means of a

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wave analyzer tuned to the desired audio output frequency. The gain depression shall not reduce the desired output by more than 8 dB.

4.5.8.3 Spurious responses. Each primary frequency band shall be completely explored for evidence of the types of spurious responses shown in a through e below which fall outside of the allowable limits as indicated in 3.14.3.

a. Internally generated spurious outputs (AGC off).

b. Image rejection. Measurements shall be made at the nominal end frequencies of each primary band, for all mixer images.

c. IF rejection. Measurements shall be made at the nominal end frequencies of each primary band.

d. Other single-signal spurious responses.

e. Front-end IM for each frequency band.

4.5.8.3.1 Internal spurious responses. Internally generated spurious responses shall be measured in USB and LSB modes for equipment operation from both internal and external frequency standards. Initial measurements shall be made in the USB mode with equipment operation from the internal frequency standard. The equipment antenna input terminals shall be terminated with 50 ohms, AGC turned off. The equipment shall be tuned to 2.0990 MHz and the USB output signal monitored with an audio spectrum analyzer and earphones. The equipment shall be tuned throughout the complete frequency range in increments of not more than 100 kHz with dial settings of xx.xx90 MHz. Spurious responses detected shall be evaluated against the requirements of 3.14.3.1.1 by substitution of a standard signal generator for the 50 ohm input termination, offsetting the generator frequency to permit amplitude comparison of the internal spurious response and that due to the applied signal, using the audio spectrum analyzer. Measurements shall be repeated for LSB operation, using dial setting of xx.xx10 MHz.

4.5.8.4 Audio output measurements.

4.5.8.4.1 Audio output voltage regulation. Measurements of audio output variation with change of load impedance and with variation of input signal level shall be made at the line output of the audio amplifier in USB and LSB. Tests shall be performed to ensure that no damage to output transistors occurs when operated into short and open circuit loads under zero drive and maximum drive conditions to verify compliance with 3.14.4.1.

4.5.8.4.2 Maximum undistorted output. Measurements shall be made in the AM, USB, and LSB modes at a nominal RF input frequency of 2.1 MHz. The AGC shall be operative, and measurements shall be made for all positions of the AGC time constant selector switch. Measurements shall be made on both rear-mounted line output receptacles and panel-mounted head telephone jacks. AF output data shall be measured for applied RF signal inputs from 1000 µV up to 1 V, with input level set in decade steps. Measurements shall be made at audio frequencies of 300, 1000, and 3500 Hz.

4.5.8.4.3 Range and linearity of audio level controls. Measurements shall be made of line output and head telephone control characteristics in the USB mode. AGC shall be active in the SLOW position of the AGC switch. Audio output as a function of angular rotation of the audio gain controls in increments of 10 degrees or less shall be tested starting from the full CCW position. The measurements shall be performed at a nominal operating frequency of 2.1000 MHz.

4.5.8.4.4 Audio level meter. Measurements shall be made to verify compliance with 3.14.4.3.1.

4.5.8.4.5 Output hum and noise. Measurements shall be made for AM, USB, and LSB modes at RF input frequencies of 2.1, 17.1, and 29.9 MHz. The input signal level shall be 1000 µV, through the antenna connector. For SSB modes, the signal shall be CW; for AM, the signal shall be modulated 90 percent at 1000 Hz. Measurements shall be made for power line frequencies of 60 and 400 Hz, as stated in a and b below.

a. Set equipment RF gain to maximum and adjust line level control to produce standard output of 1 mW. With this level as a reference, measure relative amplitude of power line related components in the output, using a wave analyzer.

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b. Set equipment line level control to maximum gain, and adjust the manual RF gain control to produce standard output of 1 mW. With this level as a reference, remove the RF input signal and measure the relative amplitude of the remaining hum and noise components with a broadband voltmeter.

4.5.8.5 Manual RF gain control. Range and linearity measurements shall be made with the equipment conditioned for USB (AGC inoperative) for an applied nominal RF input frequency of 2.1 MHz. Gain reduction in dB as a function of angular rotation in intervals of not greater than 10 degrees shall be tested.

4.5.8.6 AGC characteristics. Equipment AGC amplitude characteristics and time constants shall be measured for USB, LSB, and AM operation at the equipment low gain frequency. AGC linearity shall be verified. Measurements shall be made of maximum change in audio output level for RF inputs between that which produces standard output (S+N)/N in the SSB modes and 1 V, open circuit. No evidence of blocking or saturation of output signal shall occur.

4.5.8.6.1 AGC time constants. Separate determination of each of the AGC time constants shall be made by switching the RF input amplitude between the level which produces standard output response and various large signal levels from 100 µV to 0.1 V, open circuit, and observing the audio output.

4.5.8.6.2 Resonant overload. Measurement shall be made in the USB, LSB, and AM (90 percent modulation) modes at the equipment high gain frequency. The equipment AGC shall be disabled, and AF and RF level controls shall be set for maximum gain.

4.5.8.6.3 Time of recovery from surge inputs. Measurements shall be made in USB and LSB modes at the equipment high gain frequency.

4.5.8.7 BFO control. The output frequency range of the BFO control shall be measured at an RF input frequency of 2.1 MHz.

4.5.8.8 In-band intermodulation. Measurements shall be made at equipment high and low gain frequencies, in USB and LSB modes, and for FAST and SLOW AGC modes. RF signal frequencies shall be made with RF signal amplitudes of -37 and -13 dBm per tone, and audio output of 3 V rms per tone.

4.5.8.9 RF input circuit. Threshold of operation of the antenna input protection circuit and insertion loss in the operational condition shall be measured at the mid-frequency in each primary (MHz) band to show compliance with requirements of 3.14.9. Equipment input circuit protection shall be verified by keying drive to a suitable RF power amplifier held in a keydown condition. The power amplifier shall provide not less than 100 V open circuit in series with 50 ohms, and the protection circuit shall be operated a minimum of 10 times at each test frequency. Circuit operating threshold shall be measured at the center of each MHz band.

4.5.8.10 Accuracy of main tuning indicator calibration. Measurement shall be made at frequencies required to verify proper operation of each frequency selector in each position in order to test the requirements of 3.3.2 and 3.14.10.

4.5.8.10.1 Tuning time measurement. Measurement shall be made of the time required to switch from 2.0000 to 29.99999 MHz.

4.5.8.11 Frequency and gain stability. Frequency and gain stability shall be measured in accordance with the conditions specified in 4.5.8.11.1 through 4.5.8.11.6.

4.5.8.11.1 Sixty-minute warm-up period. The warm-up period shall begin after exposure of the equipment to the ambient temperature for a period of not less than 6 hours with no power applied, to establish the temperature at the same value as the ambient. Frequency measurements shall be made at intervals of not more than 2 minutes for the first 12 minutes, with the first measurements made 2 minutes after power is applied. Measurements shall be made at intervals of not more than 5 minutes for the next 18 minutes, and at intervals of not more than 10 minutes for the last 30 minutes.

4.5.8.11.2 Line voltage and frequency variation. Measurements shall be in accordance with the procedures specified in MIL-E-16400. A minimum stabilization period of 30 minutes shall be observed following each change of supply voltage and frequency.

4.5.8.11.3 Twenty-four-hour frequency drift. All equipment power shall be removed for a period of at least 6 hours prior to stability measurements. The equipment shall be energized for a stabilization period not to exceed 48 hours, and operated during the test period with temperature, humidity, and line voltage controlled as specified in 3.14.11.1.6.

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4.5.8.11.4 Twenty-four-hour gain stability. Gain stability shall be measured in the ISB mode (signal in both sidebands), with AGC off. Equipment conditioning periods and temperature, humidity and line voltage control, shall be as specified in 4.5.8.11.3.

4.5.8.11.5 Control variation. Measurement shall be made to verify compliance with 3.14.11.1.5.

4.5.8.11.6 Frequency standard adjustment range. Measurements shall be made to verify compliance with 3.14.14.

4.5.8.12 Reverse regeneration. Each primary frequency band shall be carefully explored for evidence of regeneration.

4.5.8.13 Oscillator radiation. Measurements shall be made at the nominal end frequencies of each primary MHz band.

4.5.8.14 Frequency standard calibration. Measurements shall be made to determine compliance with the requirements of 3.14.14 through 3.14.14.3.

4.5.8.14.1 External frequency standard. Operation using an external frequency standard shall be made to verify compliance with 3.14.14.1.

4.5.8.15 Frequency standard switch functions. Frequency standard assemblies shall be inspected to verify compliance with 3.14.14.3. This may be a qualitative test per 4.5.7.

4.5.8.16 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. Measurements of equipment gain variations (AGC off) and frequency stability shall be made to determine compliance with the requirements of 3.14.11.1.2 and 3.14.11.1.4.2. Measurements shall be made in USB, LSB, and CW modes for a minimum of one high band and one low band frequency. Frequency accuracy (synthesizer lock) checks shall be made during temperature operating tests at the end of the period of operation at 0°C, after 4 hours of operation with the equipment stabilized at +50°C, and after return to room temperature. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to 50°C by radiation-baffled heating and cooling sources as required (see 4.5.8.16.1). During humidity tests, measurements shall be made at the end of the second and fifth cycles (see 3.14.10. and 4.5.8.10).

4.5.8.16.1 Chamber temperature control provisions. The sources for maintaining the chamber environment shall be arranged or suitably baffled so that no surface at a temperature other than the desired ambient is visible to the unit under test. The maximum allowable temperature difference between any two points in the chamber shall be 10°C. The chamber control provisions shall be capable of maintaining the temperature indicated by a selected chamber ambient sensor constant within $\pm$1°C of any selected test temperature.

4.5.8.16.2 Special requirements for humidity tests. During the checks performed at the end of the second and fifth cycle of the humidity test, the equipment shall be energized for a maximum period of 30 minutes to verify proper operation. At the end of the 30 minute period, frequency for SSB modes shall be within two parts in 10^7 of the nominal frequency and maximum permissible gain change (AGC off) shall not cause audio output levels to vary by more than 8 dB (total) from those recorded as references at the end of the initial dry-out period.

4.5.8.17 Shock and vibration. Shock and vibration tests shall be performed as specified in MIL-STD-167-1 (SHIPS). Prior to and after completion of these tests, the following tests shall be performed:

a. Sensitivity measurements in accordance with 4.5.8.1 in the CW, USB and LSB modes for the 2 and 29 MHz bands only.

b. Synthesizer lock checks resulting from 4.5.8.10 testing.

c. Intermodulation distortion in accordance with 4.5.8.8. Measurements of gain and frequency stability shall be made prior to and after testing. Measurements shall be made as described in 3.14.11.

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4.5.8.18 Supply line voltage and frequency variation and transients. Tests shall be performed in accordance with the Electrical Tests section of MIL-E-16400. Reference gain, sensitivity, and frequency measurements shall be made prior to start of testing. Variation of frequency and gain with line voltage and frequency shall be measured against the requirements of 3.14.11.1. for a nominal line voltage of 115 VAC and for a nominal line frequency of 60 Hz. Following performance of the electrical tests specified in MIL-E-16400, no permanent changes shall be produced in frequency, gain, or sensitivity.

4.5.8.19 Power interruption and spike tests. Power interruption tests shall be performed in accordance with the requirements of MIL-E-16400. The 2000 and 1000 V spike tests shall be performed in accordance with the requirements of MIL-STD-1399, Section 300. Source impedance for the spike shall be 25 ohms, resistive. Spikes shall be delivered line-line and line-ground for each primary power line. For each test, a positive-going spike shall be applied at the positive crest of the supply line voltage waveform and at the zero axis crossing, and a negative-going spike shall be applied at the negative crest.

4.5.8.20 Power and power factor. Tests shall be conducted in accordance with the Power and Power Factor section of MIL-E-16400.

4.5.8.21 Chassis orientation. Verification of the ability to orient the chassis ±90 degrees shall be performed.

4.5.8.22 Assembly identification. All applicable assemblies shall be inspected to verify compliance with 3.13.1.13.

4.5.8.23 EMC. Compliance with 3.14.16 shall be verified by tests performed in accordance with MIL-STD-462. Group C inspection shall be performed for tests CE01, CE03, CS06, and RE02 and any other tests requested by the Navy for this procurement. Tests shall be performed at a power line frequency of 60 Hz.

4.5.8.24 Remote operation. All remote controlled functions shall be verified in accordance with 3.14.17 and 4.5.7.

4.5.8.25 Scan functions. Scan functions shall be verified to comply with 3.14.18.

4.5.8.26 Carrier operated relay (COR). The COR shall be verified to comply with 3.14.19. This may be a qualitative test per 4.5.7.

4.6 Standard test conditions.

4.6.1 Primary power. The nominal primary power test condition shall be 115 VAC, 60 Hz, single phase.

4.6.2 Ambient temperature. Except where otherwise specified herein, the nominal test environment temperature shall be 25°C.

4.6.3 Relative humidity. The relative humidity shall be less than or equal to 95 percent.

4.6.4 External - internal frequency standard switch. The frequency standard (FREQ STD) switch shall be set at its INT position.

4.6.5 Audio frequency (AF) output. Unless otherwise specified herein, all AF outputs shall be measured with a 600 ohm nonreactive load connected across the output terminals.

4.6.6 RF input to equipment. All RF inputs to the equipment shall be applied to the antenna input terminals through a standard dummy antenna.

4.6.7 Equipment beat frequency oscillator (BFO) control. Normally, the BFO control shall be adjusted at its zero or mid-scale position, but may otherwise be adjusted, as required in 3.14.7, to determine the performance of the equipment.

4.6.8 Scan mode. During the testing of the above-mentioned basic receiver functions, the scan mode shall be off. Performance testing shall then be conducted as shown in 4.7.3.

4.6.9 Local/remote operation. The Local/Remote Switch shall be in the local mode.

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4.7 Reliability acceptance test. A reliability acceptance test shall be performed in accordance with Test Plan IV C of MIL-STD-781. Four units from the first month's production shall comprise the test lot. Environmental conditions, operating cycles, and performance shall be as specified herein.

4.7.1 Test failure. In addition to the failure definition of 6.4.15, any parameter deviation which requires adjustment or alignment of the equipment under test shall be classified as a failure. Receiver failures occurring during the test shall be handled in accordance with the Failure Actions paragraph of MIL-STD-781.

4.7.2 Environmental conditions for reliability testing. The following environmental conditions shall be maintained throughout the reliability acceptance testing:

a. Temperature in the test chamber shall be cycled between 0°C ±2°C and 50°C ±2°C in accordance with Figure 4. The temperature rate of change shall be not less than 5°C per minute.

b. A 4 to 33 Hz sine sweep vibration shall be applied continuously in a single axis at the amplitude of 0.020 inch (double amplitude (DA)) ±0.004 inch DA. The sweep time shall be 10 minutes (up and down).

c. Input voltage shall be 115 VAC, 60 Hz. Input voltage shall be cycled in accordance with the following 3 day cycle:

(1) Nominal
(2) Nominal +10 percent
(3) Nominal -10 percent.

d. Relative humidity: Ambient.

4.7.3 Performance tests. At least daily performance measurements shall be made in accordance with the tests described below. The receiver shall store in memory the test frequencies, modes, and IF bandwidths shown in Table VI. AGC threshold shall be set at minimum and scan dwell shall be set to maximum. The receiver shall be set to the internal frequency standard.

a. Scan/lock: The receiver shall scan and lock at the frequencies shown in Table VI.

b. Gain and sensitivity: Receiver sensitivity and gain shall meet the requirements of 3.14.1.1 and 3.14.1.2, respectively, for all six modes of operation.

c. Frequency stability: The drift of the internal frequency standard shall not be greater than 5 parts in 10^8 per day, nor 3 parts in 10^7 per month.

Upon completion of the above performance tests, the receiver shall be placed in the scan mode using, in daily rotation, the test frequencies and operational modes of Table VI.

TABLE VI. Test Modes and Frequencies.

	Mode	Test Freq. (MHz)	Bandwidth (kHz)
Channel 1.	AM	2.0015	6
2.	FM	3.2750	16
3.	CW	5.3340	0.3
4.	ISB	9.5000	3
5.	USB	16.0015	3
6.	LSB	29.9909	3

4.8 Maintainability demonstration.

4.8.1 First Article Test maintainability demonstration. As a part of the First Article Test, a maintainability demonstration shall be performed at the organizational level. The demonstration shall be performed by qualified technicians and shall be used to verify that the equipment MTTR requirement is fulfilled.

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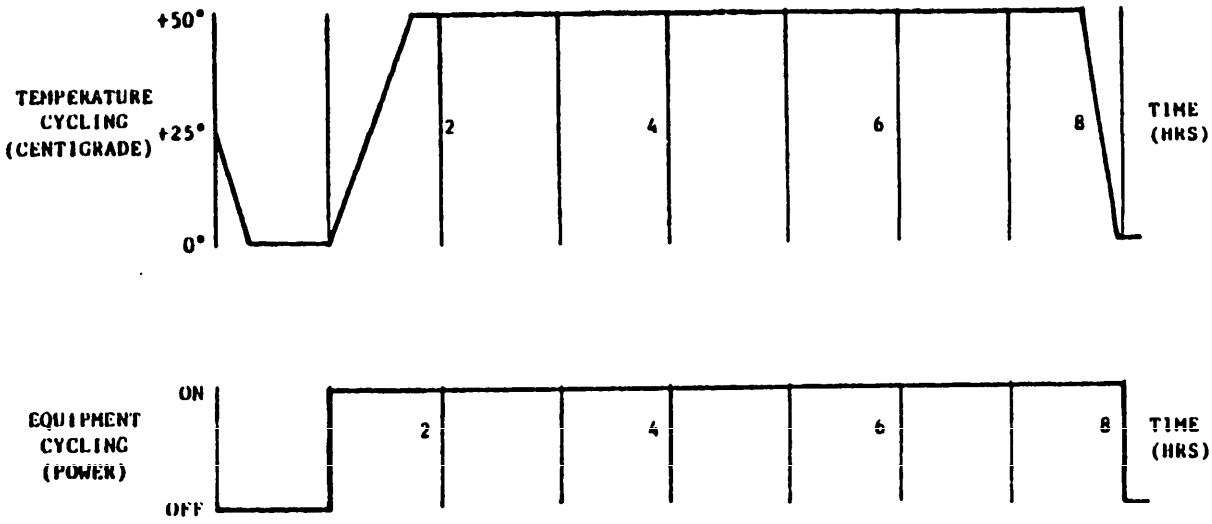


FIGURE 4. Reliability temperature and equipment cycling

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4.8.1.1 Demonstration conditions. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 by replacement of subassemblies (modules, individual PCBs) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. One-hundred-candidate corrective maintenance faults shall be determined in accordance with Appendix A of MIL-STD-471. The following information shall be made available for each candidate task:

a. Designation of specification faulty part

b. Failure Mode

c. Means of introducing fault (substitution of faulty part or simulation thereof).

The procuring activity, or its authorized representative, shall use the candidate tasks as a guide to select a sample of 50 tasks for the demonstration.

4.8.2 Accept or reject criteria.

4.8.2.1 Corrective maintenance. The accept or reject criteria for the demonstration of the corrective maintenance times for the demonstration shall be as indicated in Table VII. Acceptance shall occur when the number of observed corrective maintenance task times which exceed the required value of each specified index, MTTR or M_{maxct} is less than or equal to that shown in the acceptance table corresponding to each index for the specified confidence level. The duration of each task shall be compared to the required value(s) and recorded as greater than or less than each index. The actual value shall also be recorded. An accept decision can be made only when an accept decision is made for both MTTR and M_{maxct}. Accept or reject criteria for preventive maintenance shall be the capability to perform each preventive maintenance task with no degradation of system performance. If degradation of system operations exists, the preventive maintenance test shall have resulted in a reject decision. More than one erroneous failure indication shall constitute failure of the demonstration.

TABLE VII. Acceptance criteria.

Criterion	MTTR	M_{maxct}
Acceptance Level	20	0
Sample Size	50	50

4.8.3 Preventive maintenance. The demonstration shall qualitatively demonstrate that on-line preventive maintenance for the equipment shall not interfere with on-line operation.

4.8.4 Demonstration environment. The demonstration at the organizational level shall be performed in an environment similar to that in which the equipment will be installed.

4.8.5 Technicians. The Government reserves the right to provide and select technicians to perform the maintainability demonstration.

4.8.6 Technical documentation. Technical documentation shall include the technical manual and related maintenance documentation to be delivered with the equipment.

4.8.7 Rejection of demonstration. Failure to conform to any of the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the contracting officer shall be immediately notified. The contractor shall perform the following:

a. Develop an approach for redesign or correction of all deficiencies.

b. Upon approval of an approach, repeat the demonstration until an accept decision is reached.

4.8.8 Acceptance criteria. The maintainability demonstration shall demonstrate that the equipment conforms to the acceptance criteria prior to any delivery.

4.9 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.6).

6. NOTES

6.1 Intended use. The equipment is intended for general purpose voice, CW, FSK, AM, and LSB/USB FM and may be used for surface ship, submarine, and shore installations.

6.1.1 Mission profile. The mission profile for this equipment is continuous operation while underway. A 15-year life is planned unless the prediction and tests indicate otherwise.

6.2 First article. When a first article is required, it shall be tested and approved in accordance with the requirements of this specification. The first article should be a first production item. The first article may consist of multiple units in order to meet test schedule requirements. The contracting officer should include specific instructions in all procurement documents, regarding arrangements for examinations, tests, and approval of the first article.

6.3 Environmental inspection. Approval to ship may be withheld at the discretion of the Government pending the decision by the contracting officer on the adequacy of corrective action taken by the contractor.

6.4 Standards and definitions.

6.4.1 Standard dummy antenna. The standard dummy antenna is a device whose electrical constants are such as to simulate the impedance characteristics of antennas suitable for use with each equipment throughout its frequency range.

6.4.1.1 Dummy antenna characteristics. A device conforming to the requirement of 6.4.1 for this equipment is a 50 ohm nonreactive resistor. This resistance includes the output resistance of the standard signal generator. If necessary, resistance shall be added externally to the standard signal generator to produce a total of 50 ohms.

6.4.1.2 Application and intent. The intent of 6.4.1.1 is that the standard signal generator produce the equivalent of the signal voltage induced by an electromagnetic field in an antenna which has the constants that are simulated by the dummy antenna. Hence, the voltage reading obtained from the standard signal generator is the open circuit voltage in series with the dummy antenna. This definition applies to all voltages specified relative to the equipment antenna input terminals. It should be noted that many standard signal generators do not indicate open circuit voltage, but rather the voltage which appears across a matched load. For these generators, the desired open circuit voltage is equal to twice the indicated output.

6.4.2 Standard output (S+N)/N ratio. Standard output (S+N)/N is 10 dB. This ratio shall be determined in terms of rms values.

6.4.2.1 CW and sideband reception modes. For CW and sideband reception modes, the output (S+N)/N ratio is the ratio between the audio output resulting from an unmodulated RF input signal producing the standard beat note frequency and the output due to noise when the RF input signal is removed.

6.4.2.2 AM reception. For AM reception, the output (S+N)/N ratio is the ratio between the audio output resulting from the RF input signal modulated 30 percent at the standard modulation frequency and the output due to noise when the modulation is removed from the RF carrier.

6.4.3 Standard audio output. Standard audio output is a nominal 6 mW dissipated into a 600 ohm nonreactive resistance across the AF line output terminals.

6.4.4 Standard gain. Standard gain is that gain which produces standard output (S+N)/N ratio (see 6.4.2).

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6.4.5 Standard modulation frequency. Standard modulation frequency is 1000 Hz.

6.4.6 Standard beat note frequency. Standard beat note frequency for CW and SSB reception is 1000 Hz.

6.4.7 Sensitivity. Sensitivity of the equipment is that characteristic which determines the minimum strength of an input signal capable of producing a desired output (S+N)/N ratio.

6.4.7.1 Standard CW sensitivity. Standard CW sensitivity is the number of µV (open circuit) of unmodulated signal from a 50 ohm standard dummy antenna required to produce a standard output (S+N)/N ratio (see 6.4.2.1).

6.4.7.2 Standard AM sensitivity. Standard AM sensitivity is the number of µV of signal from a 50 ohm standard dummy antenna, modulated 30 percent, required to produce a standard output (S+N)/N ratio (see 6.4.2.2).

6.4.8 Selectivity. Selectivity of the equipment is that characteristic which determines the extent to which it is capable of discriminating between a desired RF signal and any undesired signals at frequencies required by the desired signal information spectrum but outside the bandwidth.

6.4.8.1 Antenna input to detector input. Equipment RF and IF selectivity is determined primarily by the degree to which the information band filters are capable of rejecting undesired signals at frequencies immediately adjacent to the desired signal information spectrum. Rejection of undesired signals is expressed in terms of dB with respect to the peak of the desired signal passband ripple.

6.4.8.2 Overall passband (antenna input/audio output). Selectivity is determined by the combined effects of all frequency selective circuits in the entire signal path.

6.4.8.3 Overload selectivity. Overload selectivity (desensitization) is the degree to which the RF amplifiers of the equipment are capable of rejecting undesired signals at frequencies outside the passband of the RF circuitry. Protection of the equipment front end signal path circuits is required to avoid any decrease in a desired signal (S+N)/N ratio by a strong undesired signal.

6.4.9 Spurious responses. In general, a spurious response in an equipment is the presence of any signal in the bandpass of the IF amplifiers which is not a function of the desired signal alone. There are several specific classes of spurious response:

a. Internally generated spurious outputs caused solely by the presence of the oscillators inherent to the equipment.

b. Signals which require the presence of a single strong signal external to the equipment; specific types are IF and image responses.

c. Signals resulting from the intermodulation of two strong external signals in the first stages of the receiver prior to the IF amplifier.

d. The superimposition of modulation from a single strong signal external to the equipment on a desired signal by a cross modulation process.

6.4.9.1 Image rejection. Images are the undesired second order products of each conversion, resulting from a signal on the opposite side of the injection frequency from the desired response. Image rejection is expressed as the ratio of input signal amplitudes, in dB at the image and resonant frequencies, for standard (S+N)/N ratio.

6.4.9.2 Intermediate frequency (IF) rejection. IF rejection is defined as the degree to which the equipment is capable of rejecting signals having the frequency(ies) of the IF amplifier(s). IF rejection is expressed as the ratio of input signal amplitudes, in dB, at the IF and resonant frequencies, for standard output (S+N)/N ratio.

6.4.10 Resonant overload. Resonant overload output is the output at which the overall gain (with NO AGC) of the equipment, as conditioned for standard CW sensitivity, falls 3 dB with an increase of input carrier level.

6.4.11 Maximum undistorted output. Maximum undistorted power output is the output power level at which the total output harmonic content is 5 percent, in voltage, for an input carrier adjusted for the standard sideband measurement frequency (see 6.4.12) at the level of 1000 µV.

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6.4.11.1 Total harmonic content. The total harmonic content is determined by measuring the residual voltages with the distortion analyzer after eliminating the fundamental of a sinusoidal modulating frequency.

6.4.12 Standard sideband measurement frequency. Standard sideband measurement frequency is 1000 Hz higher than the carrier for the USB and 1000 Hz lower than the carrier for the LSB.

6.4.13 AGC time constants.

a. Attack time: The time from application of the large RF signal level until the audio output amplitude has stabilized within 3 dB of the final steady-state output

b. Decay time: The time after removal of the large RF signal, for the audio output voltage to rise from 10 to 63 percent of the final value due to the small signal.

6.4.14 AGC threshold level. AGC threshold level is the RF input voltage for which the RF input to AF output characteristic departs from linearity by 1 dB.

6.4.15 Test failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the specified operational requirements.

6.5 Maintenance concept. The equipment is part of a family of HF communications equipments which are designed to minimize equipment down time through the use of plug-in assemblies. Combined with the maintenance complexity resulting from high density packaging, this results in a maintenance program entailing organizational, intermediate, and depot level maintenance.

6.6 Ordering data.

6.6.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 6.2
c. When reinspected production control and environmental test units may not be accepted on the contract (see 4.4.4)
d. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.6.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the follow paragraphs:

Paragraph	Data Requirement	Applicable DID
4.7.1	Corrective Action Plan	DI-R-7038
4.7	Reliability Test Plan	DI-R-7033/M
4.7	Reliability Test and Demonstration Reports	DI-R-7034
4.7	Reliability Test Procedures	DI-R-7035/M
4.8	Maintainability Demonstration Test Plan	DI-R-2129A
4.3, 6.2	First Article and Quality Conformance Inspection Test Procedures	UDI-T-22710B
4.3, 4.4	First Article and Quality Conformance Inspection Test Report	DI-T-5247

Preparing activity:
Navy (EC)
(5820-N776(EC))

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	☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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